

The Second Referee Pass over the v12 Proof

Proposition 6.16: The cup square on the flag quotient — a hand derivation

Instruments manuscript v12, source lines 1857–2138, with the integration points:

Lemma 6.17, Remarks 6.18–6.19, the statement, the open-problem list,
and the main-article sync clauses

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Charge and method. The full seven-step proof was re-read line by line and re-derived by hand a second time, with the repairs in place; this reading was explicitly a *seam* reading — each repaired passage was checked in context, against its neighbours, for dangling dependencies, contradictions with unrepaired text, and claims silently changed by the patches. Every checkable arithmetic claim was re-computed by a fresh exact-integer script (`w33_second_pass_check.py`, **178 checks, all PASS**, exit 0) written from scratch for this pass, independently of the manuscript, of the machine data, and of the w30/w31/w32 verifier scripts. Four bugs in the checker’s own code were caught and fixed en route (a shadowed function name, an orbit-basis ordering assumption, a wrong kernel-basis convention in the Smith-normal-form routine, and a degree mismatch in the σ_2 -bookkeeping evaluation); each had produced spurious failures against mathematics that re-derivation confirmed was correct.

1. Verdict, up front

The repairs read cleanly end-to-end. The v12 proof is now a coherent, self-consistent, line-checkable derivation from Step 0 to Step 7, and the referee cycle of Waves 30–32 is closed. All eleven repair hunks are present (R1’s five from Wave 31; R2–R7’s six from Wave 32); all seven superseded phrasings are gone; every load-bearing claim — including every claim the repairs introduced — was independently re-verified, and the cross-step dependencies that the repairs created or relied on all resolve.

No mandatory corrections. Two residual polish items are recorded below (Findings F1 and F2): each is a single optional clause, neither affects any assertion of the Proposition, and neither rises to the level of a repair. In journal terms: *accept; the minor revision has been executed and executed well*.

The epistemic chain on the Proposition now runs: hand-derived (W29), machine-certified (13C), independently re-implemented (W28), referee-verified (W30, findings R1–R7), lattice-gated (W31, R1 applied plus the G3L gate), referee-complete (W32, R2–R7 adjudicated and applied), and now **second-pass confirmed** (W33, this wave).

2. What the second pass confirmed

- **R1, all five hunks, seam-clean.** The statement clause ($H^6(B; \mathbb{Z}) \cong \mathbb{Z}$, the discriminant twice a generator) is correct and *well defined*: the τ^3 -tail ambiguity of the Δ -lift in $H^6(\text{BK})$ dies in the quotient, so the image of Δ in $H^6(B)$ is lift-independent — a dedicated gate (M4, lift-independence for $c = 0, 1, 2$). Step 2’s lattice layer re-verified: the index table 1, 1, 1, 2, 2, 4, 8 for degrees 0–6 (determinants of the change-of-basis), the four polynomial identities, the witness $S_1 = \frac{1}{2}(\sigma_1\sigma_2 - 3\sigma_3 + \Delta)$ non-integral with $2S_1$ integral. Step 3’s display $\mathbb{Z}\{A_1, S_1, S_2, \sigma_3\} \oplus \mathbb{Z}/3 \cdot \tau^3$ with the index-two note: exact. Step 7’s quotient, run on the orbit-sum lattice,

$$(\mathbb{Z}\{A_1, S_1, S_2, \sigma_3\} \oplus \mathbb{Z}/3 \cdot \tau^3) / \langle \sigma_1^3, \sigma_1\sigma_2, 2\tau^3, \sigma_3 \rangle \cong \mathbb{Z} \cdot \langle S_1 \rangle,$$

is re-computed by Smith normal form in the paper’s coordinates — free rank 1, no torsion, the membership triple ($\Delta - 2S_1$ a relation; $\Delta - S_1$ and S_1 not) exact, $\Delta = 2S_1$. The remark

knock-on is present and now accurate: the G3L gate certifies exactly what the residuals list names.

- **R2, the corrected two-step, mechanism complete.** Odd r : targets in odd p -degree, the forward reference marked and *supported* by Step 3’s closing sentence (line 2024); gate M2 re-proves $\ker N \subseteq (T - 1)M$ lattice-exactly for $d \leq 8$. Even $r \geq 8$: source fibre-degrees exactly $\{8, 9\}$; the degree-9 classes die under d_2 (free) or d_4 (torsion, $2\tau^{a+2} \neq 0$); the degree-8 σ_1 -multiples are d_2 -boundaries; the rest die under $d_4(\beta z_3 z_5) = \beta(\sigma_2 + 2\tau^2)z_5 \neq 0$ on the E_4 page — the exact point the referee’s own original sketch missed. The σ_1 -divisibility lattice fact $\{\beta \in M_d : \sigma_1 \mid \beta\} = \sigma_1 M_{d-1}$ is re-verified for degrees 1–8 (rank match plus index 1, degree by degree), and the σ_2 -side bookkeeping for degrees 2–8 (σ_2 at the locus $\chi_3 = -\chi_1 - \chi_2$ is $-(x^2 + xy + y^2) \neq 0$).
- **R3, the (2, 2)-clause.** The Step 5 enumeration is now genuinely complete: all five bidegrees of total degree four are accounted for, the vanishing $E_2^{2,2} = 0$ resting on the exterior algebra having no degree-two class (M5).
- **R4, both hunks, load-bearing and supplied.** The display $\iota^* c_1 = \sigma_1 + b\tau$ with the honest parenthetical, and the pin $0 = s^*(\iota^* c_1) = s^*(\sigma_1 + b\tau) = bu$, so $b = 0$: the regular representation’s Chern classes re-computed in $\mathbb{Z}[u]/(3u)$ ($(1+u)(1+2u) = 1+3u+2u^2$), the pins $(b, \varepsilon, \varepsilon') = (0, 2, 0)$ forced as printed. The consequence is real: with b pinned, $d_2(\tau z_1) = \tau\sigma_1 = 0$ is line-checkable and Step 5’s incoming-image completeness rests on an established fact; unpinned, the crux dies — the counterfactual quotient re-computed and confirmed (M3: the trivial group).
- **R5, the two-step quotient.** Logically valid as scoped: after the direct-summand quotient there is exactly one relation, and the order-three argument applies to it. The sum-quotient re-verified ($\mathbb{Z}/3$, $[\tau^2]$ of order three generating, $[\sigma_2] = -2[\tau^2]$), the logical-gap witness re-exhibited, the old per-subgroup sentence gone.
- **R6, the convention parenthetical.** Present at the identification sentence, correct (the sign squares away, $(\pm 1)^2 = 1$), and it does not interfere with the functorial part $\kappa = \pi \circ \gamma$.
- **R7, the $\tau\Delta$ marking.** Accurate in both directions: the leading-level claim is correct, the three lift-values are distinct, and the downstream uses really do avoid it — the σ_i -instances are correctly marked unconditional via Step 4’s pins.

3. Findings (optional; one clause each; no mathematics affected)

F1 (nano, presentational). Step 1’s repaired text uses σ_1 , σ_2 , τ — and the transgression value $\sigma_2 + 2\tau^2$ that Step 4 pins — with first occurrences at source lines 1931/1933, while σ_i is introduced at line 1969, τ at line 1983, and the value pinned at line 2047: thirty-eight lines of forward dependency. The odd- r clause carries its marker (“by Step 3 below”); the even- r clause carries none. There is no logical circularity (Steps 2–4 are established by independent means), but a linear reader meets the symbols before their definitions; one parenthetical would close it. In the same sentence, the lead-in “the fibre classes there are *boundaries* before the E_8 -page” conflates two death modes — the σ_1 -multiple degree-8 classes are d_2 -boundaries, the rest die as *non-cycles*; the elaborating sentences are precise about exactly this, so only the lead-in word is loose (“killed” for “boundaries”).

F2 (nano, presentational). Step 3’s display presents the orbit sums S_1, S_2 as named classes, but they are defined only up to τ^3 -tails (the display chooses a splitting; only A_1 and the pinned σ_i are canonical). Step 7’s quotient computation is nonetheless **lift-independent** — the relation $2\tau^3 = 0$ kills the ambiguity in the quotient, verified as a dedicated gate (M4) — and this is exactly what makes the statement’s “discriminant class” well defined in $H^6(B; \mathbb{Z})$

despite the R7-marked ambiguity in $H^6(\text{BK})$. The paper never says so explicitly; half a clause in Step 7 would close it.

Neither finding is a defect in a claim; both are one-clause polish opportunities of the same species as R7 was (a marking), strictly weaker: nothing downstream depends on them.

4. The integration points (re-assessed, all clean)

Lemma 6.17 is unchanged (degrees ≤ 4 only) and its references to the proposition and the machine certificate remain accurate. Remark 6.19’s residuals list names the lattice layer and the twice-a-generator position, which G3L genuinely certifies; the phrase “a self-contained invariant-theory step” now points at a correct, self-contained step. Remark 6.18’s differentiation clause remains accurate. The statement is internally consistent with Step 7, and the “discriminant class” is well defined in $H^6(B)$ by the lift-independence gate. The main article correctly stays at v10: its sync clauses are prose-level and contain no proof internals (verified), so nothing in the v12 repairs creates a drift. The open-problem (vi) discharge is unaffected.

5. Referee’s summary

The first pass asked whether the proof was *correct*; this pass asked whether the *repaired* proof *reads* as a proof — whether the eleven surgical hunks, each verified in isolation by their own waves, compose into a text a reader can walk through without tripping over a seam. They do. The forward-dependency structure is sound (and where it exists it is either marked or, per F1, one clause short of being marked); the load-bearing dependency created by the repairs (Step 5’s completeness resting on the R4 pin) is supplied rather than silent; the superseded statements are gone everywhere; the corrected top-degree story is told consistently in the statement, Steps 2/3/7, and the certificate remark; and the two independent verification layers (W31’s G3L, W32’s twelve gates) are now joined by this pass’s 178 checks, which re-derive the whole chain from the orbit decomposition up. The residue — two optional clauses — is the honest size of what remains.

The mathematics, final state: $x^2 \neq 0$ in $H^4(B; \mathbb{Z})$; x generating H^2 and x^2 generating H^4 ; the additive tuple $(\mathbb{Z}, 0, \mathbb{Z}/3, \mathbb{Z}/3, \mathbb{Z}/3, \mathbb{Z}/3, \mathbb{Z})$; and $H^6(B; \mathbb{Z}) \cong \mathbb{Z}$ with the discriminant twice a generator — hand-derived, machine-certified, independently re-implemented, referee-verified, lattice-gated, referee-complete, and now second-pass confirmed. `thm:state-d2` and $\delta_2(\mathcal{D}(\mathbb{C}^3)) = 4/3$ untouched; the ququart bracket $[4/3, 3/2]$ open; the W23 stop stands.

Overall recommendation: accept — the revision requested by Wave 30 has been executed (W31: R1; W32: R2–R7, adjudicated) and survives a full second read; the two polish items F1/F2 are optional.

6. Artifacts

- `w33_second_pass_check.py` — the independent exact-integer verifier (13 gate families, 178 checks: the seam/textual layer P1–P5 and the mathematical layer M1–M8, including the new lift-independence gate M4 and the σ_1/σ_2 -bookkeeping gates M5).
- `w33_second_pass_output.txt` + `w33_second_pass_check.json` — the transcript and the structured record (all PASS, exit 0), with the honest debug trail of the four checker-code bugs caught en route.
- `WAVE33_SECOND_REFEREE_PASS.md` — the wave note; this document is its LaTeX rendition.